Calculus II 2017 Final Solutions

1. Evaluate the iterated integral $\int_0^3 \int_0^1 x\sqrt{x^2+y}\,\mathrm{d}x\,\mathrm{d}y$.

Solution

$$\begin{aligned}
\int_0^3 \int_0^1 x\sqrt{x^2+y}\,\mathrm{d}x\,\mathrm{d}y &= \int_0^3 [\frac{2}{3}\cdot\frac{1}{2}(x^2+y)^{3/2}]_0^1 \\
&= \frac{1}{3}\int_0^3 (1+y)^{3/2} - y^{3/2}\,\mathrm{d}y \\
&= \frac{1}{3}\frac{2}{5}[(1+y)^{5/2} - y^{5/2}]_0^3 = \boxed{\frac{2}{15}(31-\sqrt{3}^5)}.
\end{aligned}$$

2. Using an integral of two variables, find the volume of the region enclosed by the parabaloid $z = \sqrt{x^2+y^2}$ and the sphere $x^2+y^2+z^2 = 1$.

Solution

In polar coordinates, this is the region enclosed by $z = r$ and (using the top of the sphere) $z = \sqrt{1-r^2}$. Solving $r = \sqrt{1-r^2}$ for r gives $r = \sqrt{1/2}$ so the domain is $0 \le \theta \le 2\pi$ and $0 \le r \le 1/\sqrt{2}$.
We then evaluate

$$\begin{aligned}
Vol &= \int_0^{2\pi}\int_0^{1/\sqrt{2}} (\sqrt{1-r^2} - r)r\,\mathrm{d}r\,\mathrm{d}\theta \\
&= 2\pi\left(\int_0^{1/\sqrt{2}} r(1-r^2)^{1/2}\,\mathrm{d}r - \int_0^{1/\sqrt{2}} r^2\,\mathrm{d}r\right) \\
&= 2\pi\left[(1-r^2)^{3/2}\cdot\frac{2}{3}\cdot(1\frac{1}{2}) - \frac{r^3}{3}\right]_0^{1/\sqrt{2}} \\
&= \frac{2\pi}{3}\left((1-\frac{1}{2\sqrt{2}}) - (\frac{1}{2\sqrt{2}})\right) = \boxed{\frac{2\pi}{3} - \frac{\sqrt{2}\pi}{3}}
\end{aligned}$$

Or you could have evaluated $\int_0^{2\pi}\int_0^{\pi/4}\int_0^1 \rho^2 \sin\phi\,\mathrm{d}\rho\,\mathrm{d}\phi\,\mathrm{d}\theta$ which turned out to be a fair bit easier.

3. Using the transformation $x = 2u$ and $y = 3v$ evaluate the integral $\iint_R x^2 \, \mathrm{d}A$ for the region $R = \{(x, y) \mid 9x^2 + 4y^2 \leq 36\}$.

Solution

The transformed region is $D = \{(u, v) \mid u^2 + v^2 \leq 1\}$ and the jacobian is

$$\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 2 & 0 \\ 0 & 3 \end{vmatrix} = 6$$

so the integral is

$$\iint_D 4u^2 |6| \, \mathrm{d}A = 24 \iint_D u^2 \, \mathrm{d}A.$$

For this, we transform again to polar coordinates, getting

$$\begin{aligned} &= 24 \int_0^1 \int_0^{2\pi} (r\cos\theta)^2 r \, \mathrm{d}\theta \, \mathrm{d}r = 24 \int_0^1 r^3 \, \mathrm{d}r \int_0^{2\pi} \cos^2\theta \, \mathrm{d}\theta \\ &= 6 \int_0^{2\pi} \frac{1}{2}(1 + \overbrace{\cancel{\cos 2\theta}}^{\int = 0}) \, \mathrm{d}\theta = \boxed{6\pi} \end{aligned}$$

4. Write (but do not evaluate) the following integral in spherical coordinates:

$$\int_{-3}^{3} \int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}} \int_{-\sqrt{9-x^2-y^2}}^{\sqrt{9-x^2-y^2}} \sqrt{x^2 + y^2 + z^2} \, \mathrm{d}z \, \mathrm{d}x \, \mathrm{d}y$$

Solution

$$\int_0^3 \int_0^{\pi} \int_0^{2\pi} \rho^3 \sin\phi \, \mathrm{d}\theta \, \mathrm{d}\phi \, \mathrm{d}\rho$$

5. Find the volume of the region enclosed between the paraboloid $x = y^2 + z^2$ and the plane $x = 16$.

Solution

$$\int_0^{2\pi} \int_0^4 \int_{r^2}^{16} r \, \mathrm{d}x \, \mathrm{d}r \, \mathrm{d}\theta = 2\pi \int_0^4 r(16 - r^2) = 2\pi(128 - 64) = \boxed{128\pi}$$

6. Evaluate the line integral $\int_C \frac{e^{-z}}{x^2+y^2} \, \mathrm{d}s$, where C is $\mathbf{r}(t) = \langle 2\cos t, 2\sin t, t \rangle$ for t from 0 to 2π.

Solution

$$
\begin{aligned}
&= \int_0^{2\pi} \frac{e^{-t}}{4\cos^2 t + 4\sin^2 t} |\mathbf{r}'(t)| \, \mathrm{d}t \\
&= \int_0^{2\pi} \frac{e^{-t}}{4} \sqrt{1 + 4\sin^2 t + 4\cos^2 t} \, \mathrm{d}t \\
&= \frac{\sqrt{5}}{4} \int_0^{2\pi} e^{-t} \, \mathrm{d}t = \boxed{\frac{\sqrt{5}}{4}(1 - e^{-2\pi})}
\end{aligned}
$$

7. Use Green's theorem to evaluate $\int_C x^2 y \, \mathrm{d}x - y^2 x \, \mathrm{d}y$ where C is the curve that goes counterclockwise around first quadrant of the circle $x^2 + y^2 = 4$.

Solution

By Green's theorem we get

$$
\begin{aligned}
\int_C x^2 y \, \mathrm{d}x - y^2 x \, \mathrm{d}y &= \iint_R \frac{\partial}{\partial x}(-y^2 x) - \frac{\partial}{\partial y}(x^2 y) \, \mathrm{d}A \\
&= \iint_R -x^2 - y^2 \, \mathrm{d}A = -\int_0^2 \int_0^{\pi/2} r^2 r \, \mathrm{d}\theta \, \mathrm{d}r \\
&= -\pi/2(2^4/4) = \boxed{-2\pi}
\end{aligned}
$$

8. Setup, but do not evaluate, an integral describing the flux of $\mathbf{F}(x, y, z) = \langle z, y, x \rangle$ through the sphere $x^2 + y^2 + z^2 = 1$.

Solution

Parametrising S by $\mathbf{R}(\theta,\phi) = \langle \sin\phi\cos\theta, \sin\phi\sin\theta, \cos\phi\rangle$ we get

$$\begin{aligned}
&\mathbf{r}_\theta \times \mathbf{r}_\phi \\
&\quad = \langle -\sin\phi\sin\theta, \sin\phi\cos\theta, 0\rangle \times \langle \cos\phi\cos\theta, \cos\phi\sin\theta, -\sin\phi\rangle \\
&\quad = \langle -\sin^2\phi\cos\theta, -\sin^2\phi\sin\theta, -\sin\phi\cos\phi\rangle
\end{aligned}$$

So

$$\begin{aligned}
\mathbf{F}\cdot(\mathbf{r}_\theta \times \mathbf{r}_\phi) &= \langle \cos\phi, \sin\phi\sin\theta, \sin\phi\cos\theta\rangle \cdot \langle -\sin^2\phi\cos\theta, -\sin^2\phi\sin\theta, -\sin\phi\cos\phi\rangle \\
&= -\sin^2\phi\cos\theta\cos\phi - \sin^3\phi\sin^2\theta - \sin^2\phi\cos\phi\cos\theta
\end{aligned}$$

The flux is

$$\begin{aligned}
\left|\iint_S \mathbf{F}\cdot \mathrm{d}\mathbf{S}\right| &= \boxed{\int_0^{2\pi}\int_0^{\pi} 2\sin^2\phi\cos\theta\cos\phi + \sin^3\phi\sin^2\theta\,\mathrm{d}\phi\,\mathrm{d}\theta} \\
&= \cancelto{0}{\int_0^{2\pi}\cos\theta\,\mathrm{d}\theta}\int_0^{\pi} 2\sin^2\phi\cos\phi\,\mathrm{d}\phi + \int_0^{2\pi}\sin^2\theta\,\mathrm{d}\theta\int_0^{\pi}\sin^3\phi\,\mathrm{d}\phi \\
&= \int_0^{2\pi}\frac{1-\cancelto{\int\cos=0}{\cos 2\theta}}{2}\,\mathrm{d}\theta\int_o^{\pi}\sin\phi(1-\cos^2\phi)\,\mathrm{d}\phi \\
&= \int_0^{2\pi} 1/2\,\mathrm{d}\theta\cdot\int_0^{\pi}\sin\phi - \sin\phi\cos^2\phi\,\mathrm{d}\phi \\
&= \pi[-\cos\phi + \frac{1}{3}\cos^3\phi]_0^{\pi} = \pi(1+1+\frac{-1-1}{3}) = \frac{4}{3}\pi
\end{aligned}$$

Or one can compute $\operatorname{div}\mathbf{F} = 0+1+0 = 1$ and so using the Divergence Theorem the integral is $\iiint 1\,\mathrm{d}V = \int_0^1\int_0^{\pi}\int_0^{2\pi}\rho^2\sin\phi\,\mathrm{d}\theta\,\mathrm{d}\phi\,\mathrm{d}\rho$.

9. Find the curl and divergence of $\mathbf{F}(x,y,z) = e^x\sin y\mathbf{i} + e^x\cos y\mathbf{j} + z\mathbf{k}$.

Solution

The curl is

$$\nabla\times\mathbf{F} = \langle 0-0, 0-0, e^x\cos y - e^x\cos y\rangle = \boxed{\langle 0,0,0\rangle}$$

and the divergence is

$$\nabla\cdot\mathbf{F} = e^x\sin y - e^x\sin y + 1 = \boxed{1}.$$

10. Find the flux of $\mathbf{F}(x,y,z) = \langle 3xy^2, xe^z, z^3\rangle$ through the surface S bounding the region E bounded by $y^2+z^2=1$, $x=-1$, and $x=2$.

Solution

We use the divergence theorem. Computing $\nabla\mathbf{F} = 3y^2 + 3z^2 = 3r^2$, we get, coordinatising in x, r and θ,

$$\begin{aligned}
\iint_S \mathbf{F}\cdot \mathrm{d}\mathbf{S} &= \iiint_E 3r^2 \cdot r \,\mathrm{d}V \\
&= \int_{-1}^{2}\int_0^1\int_0^{2\pi} 3r^3 \,\mathrm{d}\theta\,\mathrm{d}r\,\mathrm{d}x \\
&= 3(2\pi)\int_0^1 3r^3\,\mathrm{d}r = \boxed{\frac{9\pi}{2}}
\end{aligned}$$